example, WO[1], corresponding to the first input of the output node. Look at the structure in Figs. 26-5 and 26-6, and ask: how will the output (*X3*) be affected if this particular weight (*w*) is changed slightly, but everything else is kept the same? The answer is:

EQUATION 26-3
Slope of the output layer weights. This equation is written for the weight, WO[1].

$$\frac{\Delta X3}{\Delta w} = X2[1]\ \text{SLOPE}_{O}$$

where SLOPE_O is the first derivative of the output layer sigmoid, evaluated where we are operating on its curve. In other words, SLOPE_O describes how much the *output* of the sigmoid changes in response to a change in the *input* to the sigmoid. From Eq. 26-2, SLOPE_O can be calculated from the current output value of the sigmoid, X3. This calculation is shown in line 3160. In line 3170, the slope for this weight is calculated via Eq. 26-3, and stored in the variable DX3DW (i.e., $\Delta X3/\Delta w$).

Using a similar analysis, the slope for a weight on the hidden layer, such as WH[1,1], can be found by:

EQUATION 26-4
Slope of the hidden layer weights. This equation is written for the weight, WH[1,1].

$$\frac{\Delta X3}{\Delta w} = X1[1]\ \text{SLOPE}_{H1}\, WO[1]\ \text{SLOPE}_{O}$$

SLOPE_{H1} is the first derivative of the hidden layer sigmoid, evaluated where we are operating on its curve. The other values, X1[1] and WO[1], are simply constants that the weight change sees as it makes its way to the output. In lines 3070 and 3080, the slopes of the sigmoids are calculated using Eq. 26-2. The slope of the hidden layer weight, DX3DW is calculated in line 3090 via Eq. 26-4.

Now that we know the *slope* of each of the weights, we can look at how each weight is changed for the next iteration. The new value for each weight is found by taking the current weight, and adding an amount that is proportional to the slope:

EQUATION 26-5
Updating the weights. Each of the weights is adjusted by adding an amount proportional to the slope of the weight.

$$w_{new} = w_{old} + \frac{\Delta X3}{\Delta w}\ \text{ELET}\ \text{MU}$$

This calculation is carried out in line 3100 for the hidden layer, and line 3180 for the output layer. The proportionality constant consists of two factors,